US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12416764
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	Chao; Feng-Chiang et al.

Optical device and associated optical collimator structure

Abstract

The present disclosure provides an optical device including a tray with a step structure, first filters, second filters, and an optical signal router. The step structure has a first portion and a second portion laterally connected to the first portion. The first portion has a first bottom surface and a first top surface. The second portion has a second bottom surface and a second top surface. The first bottom surface and the second bottom surface are substantially coplanar, and the first portion is thinner than the second portion. The first filters are mounted on the first top surface. The second filters are mounted on the second top surface. The optical signal router optically couples to the first filters and the second filters, and is configured to receive a light beam, transmissible to the tray, from one of the first filters or the second filters.

Inventors: **Chao; Feng-Chiang (Taichung, TW), Peng; Chang-Yi (Taoyuan, TW)**

Applicant: **BROWAVE CORPORATION** (Hsinchu County, TW)

Family ID: **1000008821854**

Assignee: **BROWAVE CORPORATION (Hsinchu County, TW)**

Appl. No.: **17/965114**

Filed: **October 13, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20240126018 A1	Apr. 18, 2024

Publication Classification

Int. Cl.: **G02B6/32** (20060101); **G02B6/293** (20060101)

U.S. Cl.:

CPC **G02B6/32** (20130101); **G02B6/2938** (20130101);

Field of Classification Search

CPC: G02B (6/32); G02B (6/29)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10514507	12/2018	Zhang	N/A	G02B 27/30
2003/0108292	12/2002	Abe	385/24	G02B 6/2931
2018/0139520	12/2017	Xiao	N/A	G02B 6/29365
2021/0041645	12/2020	Li	N/A	H04B 10/506

Primary Examiner: Wong; Tina

Attorney, Agent or Firm: MUNCY, GEISSLER, OLDS & LOWE, P.C.

Background/Summary

FIELD
(1) The present disclosure relates to an optical device and associated optical collimator structure, particularly, to a single-sided MUX/DeMUX device and associated optical collimator structure.
BACKGROUND
(2) Wavelength Division Multiplexing (WDM) is a fiber-optic transmission technique that enables the use of multiple light wavelengths (or colors) to send data over the same medium. Dense WDM (DWDM) is defined in terms of frequencies. DWDM's tighter wavelength spacing fits more channels onto a single fiber, but costs more to implement and operate. DWDM is for systems with more than eight active wavelengths (or channels) per fiber. Minimizing device complexity, reducing production cost, and enhance the assembling precision through structural design is critical to DWDM manufacturing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
(2) FIG. **1** is a schematic diagram of an optical device from a top view perspective according to some embodiments of the present disclosure.
(3) FIG. **2** is a schematic diagram of an optical device from a side view perspective according to some embodiments of the present disclosure.
(4) FIG. **3** is a schematic diagram showing an optical path in a portion an optical device from a top view perspective according to some embodiments of the present disclosure.
(5) FIG. **4** is a schematic diagram showing a portion of an optical device with a beam shape

modifier according to some embodiments of the present disclosure.
(6) FIG. **5** is a schematic diagram showing a portion of an optical device with a beam shape modifier according to other embodiments of the present disclosure.
(7) FIG. **6** is a schematic diagram of an optical collimator structure from a side view perspective according to some embodiments of the present disclosure.
(8) FIG. **7**A is a schematic diagram of an optical collimator structure from a perspective traversing a longitudinal direction thereof according to some embodiments of the present disclosure.
(9) FIG. **7**B is a schematic diagram of an optical collimator structure from a perspective traversing a longitudinal direction thereof according to other embodiments of the present disclosure.

DETAILED DESCRIPTION

(10) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
(11) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
(12) Notwithstanding that the numerical ranges and parameters setting forth the broad scope of the disclosure are approximations, the numerical values set forth in the specific examples are reported as precisely as possible. Any numerical value, however, inherently contains certain errors necessarily resulting from the standard deviation found in the respective testing measurements. Also, as used herein, the terms “substantially,” “approximately,” or “about” generally means within a value or range which can be contemplated by people having ordinary skill in the art. Alternatively, the terms “substantially,” “approximately,” or “about” means within an acceptable standard error of the mean when considered by one of ordinary skill in the art. People having ordinary skill in the art can understand that the acceptable standard error may vary according to different technologies. Other than in the operating/working examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for quantities of materials, durations of times, temperatures, operating conditions, ratios of amounts, and the likes thereof disclosed herein should be understood as modified in all instances by the terms “substantially,” “approximately,” or “about.” Accordingly, unless indicated to the contrary, the numerical parameters set forth in the present disclosure and attached claims are approximations that can vary as desired. At the very least, each numerical parameter should at least be construed in light of the number of reported significant digits and by applying ordinary rounding techniques. Ranges can be expressed herein as from one endpoint to another endpoint or between two endpoints. All ranges disclosed herein are inclusive of the endpoints, unless specified otherwise.
(13) A wavelength division multiplexing (MUX) is a technique utilized to combine a number of optical signals carried by light having different wavelengths into an optical fiber (or other suitable transmission waveguide). Combined light exiting from an optical fiber (or other suitable

transmission waveguide) can be decomposed into individual channels having different wavelength bands using a demultiplexing technique (DeMUX). Alternatively stated, a device that multiplexes different wavelength channels or groups of channels into one fiber (or other suitable transmission waveguide) is a multiplexer, and a device that divides the multiplexed channels or groups of channels into individual or subgroups of channels is a demultiplexer. The optical devices implementing such MUX and DeMUX techniques are respectively referred to a multiplexing and demultiplexing module, or simply a multiplexer or a demultiplexer.
(14) Demultiplexer and multiplexer can be utilized in various types of optical devices, optical transmission device or semiconductor structures in order to convey signal in specific ways. However, more channels the multiplexer or demultiplexer are carrying, more densely packed optical fibers, or more collimators carrying such optical fibers, should be assembled on the optical device. In one of the current practices, collimators and optical fibers are placed on both sides of a suitable carrier tray in order to accommodate the large number of the collimators and optical fibers to be assembled. However, the manufacturing complexity and cost for the double-sided structure set forth prevents the wide acceptance of such model. For example, the collimators and optical fibers has to be individually assembled in a sequence corresponding to the optical path, with active calibration and alignment, rendering constant flipping of the carrier tray and laborious calibration process. In another one of the current practices, collimators and optical fibers are stacked in a multi-layer fashion on at least one side of a suitable carrier tray in order to accommodate the large number of the collimators and optical fibers to be assembled. The multi-layer stacking structure is implemented with multiple wedge components so as to support individual collimators and optical fibers, which is often in cylindrical shape and cannot provide a stable placement alone. Given the above, present disclosure provides an optical device and an optical collimator structure that cures the deficiencies of the structures used in current practice.
(15) FIG. **1** and FIG. **2** are schematic diagrams of an optical device **10** from a top view perspective (X-Y plane) and a side view perspective (X-Z plane) according to some embodiments of the present disclosure. The optical device **10** includes a tray **11**, first filters **12**, second filters **13**, an optical signal router **14**, first collimators **15**, and second collimators **16**.
(16) The tray **11** includes one or more step structure having a first portion **111** and a second portion **112** laterally connected to the first portion **111**. The first portion **111** has a top surface t**1** and a bottom surface b**1**, and the second portion **112** has a top surface t**2** and a bottom surface b**2**. As illustrated in FIG. **2**, the first portion **111** is a thinner portion of the tray **11**, and the second portion **112** is a thicker portion of the tray **11**. The bottom surface b**1** and the bottom surface b**2** are substantially coplanar. A thickness of the first portion **111** is less than a thickness of the second portion **112**, therefore the top surface t**2** is higher than the top surface t**1** in the Z-direction. In some embodiments, the entirety of the tray **11** is a continuous substrate. Depending on the number or collimators to be assembled, the number of the step structure can be increased based on various design needs.
(17) Referring to FIG. **1**, the first filters **12** and the first collimators **15** are mounted on the top surface t**1**, and the second filters **13** and the second collimators **16** are mounted on the top surface t**2**. The optical signal router **14** is coupled to a side surface of the tray **11** proximal to the second portion **112** of the tray **11**.
(18) The first collimator **15** includes a common collimator **151** and first channel collimators **152**. The optical device **10** can be operated as a multiplexer or a demultiplexer. When the optical device **10** is operated as the multiplexer, the optical device **10** is configured to receive light beams respective from the first channel collimators **152** and the second collimators **16**, and further configured to multiplex the light beams for outputting through the common collimator **151**. On the other hand, when the optical device **10** is operated as the demultiplexer, the optical device **10** is configured to receive a light beam from the common collimator **151**, and further configured to demultiplex the light beam for outputting through the first channel collimators **152** and the second

collimators **16**.

(19) The first filters **12** are optically coupled to the respective first collimators **15**, and the second filter **13** are optically coupled to the respective second collimators **16**. More specifically, the first filters **12** includes a common filter **121** and first channel filters **122**; the common filter **121** is optically coupled to the common collimator **151**, and the first channel filters **122** are optically coupled to the respective first channel collimators **152**.

(20) A pass band of respective first channel filters **122** and the second filters **13** are different from each other, and a pass band of the common filter **121** includes each of the pass bands of the respective first channel filters **122** and the second filters **13**. In some embodiments, the common filter **121** is a prism.

(21) The optical signal router **14** optically couples the first filters **12** and the second filters **13**, and is configured to receive a light beam, transmissible to the tray **11**, form one of the first filter **12** or the second filters **13**. In some embodiments, the optical signal router **14** is a right angle prism vertically offsetting the light beam from one of the first filters **12** or the second filters **13**.

(22) In FIG. **2**, take demultiplexer for an example, an optical path OP is illustrated using dash lines. When a light beam is received by the common collimators **151**, the light beam is transmitted through the common filter **121**. After transmitting through the common filter **121**, the light beam is further transmitted through the second portion **112** of the tray **11** to the optical signal router **14**. The tray can be composed of materials which are transparent or transmissible to the wavelength channels that the common collimators **151** carrying. Subsequently, the light beam is guided, for example, through multiple reflections in the optical signal router **14**, to vertically elevate its level and continue the path over the top surface t**2**.

(23) When the light beam reaches the second filter **13**, the second filter **13** transmits a portion of the light beam passes through to the corresponding second collimator **16**, and the second filter **13** reflects the remainder of light beam toward the optical signal router **14**. The remainder of the light beam is guided to one of the first filters **12**. The optical path of the remained light beam is again guided, for example, through multiple reflections in the optical signal router **14**, to vertically descend its level and enter the second portion **112** of the tray **11**, continuing the path over the top surface t**1** (not shown in FIG. **2**).

(24) Referring to FIG. **3**. FIG. **3** is a schematic diagram showing an optical path OP in a portion the optical device **10** from a top view perspective. As shown in FIG. **3**, after the light beam is received by the common collimator **151**, the light beam is vertically elevated to the second filter **13**, as previously described in FIG. **2**. The second filter **13** then transmits the portion of the light beam passes through (shown by a dotted lines) to the second collimator **16**, and the second filter **13** reflects the remainder of the light beam to the optical signal router **14**. The optical signal router **14** further vertically descended the remainder of light beam to enter the second portion **112** of the tray **11**, and continue on a path that bounds to the first channel filter **122** disposed on the top surface t**1** of the tray **11**. Similarly, when the light beam reaches the first channel filter **122**, the first channel filter **122** transmits a portion of the light beam passes through to the first channel collimator **152**, and the first channel filter **122** reflects the remainder of light beam to the optical signal router **14** by transmitting through the second portion **112** of the tray **11**. So on and so forth, the optical device **10** serving as a demultiplexer enabled the demultiplexing of multi-channel signals through a compact structure, which assembled the collimators on a single side of the tray **11**, and through the at least one step structure on a designated side of the tray **11**, the large number of collimators can be accommodated in said single-sided tray **11**. As previously discussed, assembling collimators or optical fibers all on one side of the tray streamlines the manufacturing process and reduce production cost by enabling passive assembling and without continuous flipping of the tray.

(25) It should be appreciated by persons having ordinary skills in the art that when the optical device **10** is operated as multiplexer, the light beam is transmitted in a reversed direction of the optical path OP shown in FIG. **2** and FIG. **3**.

(26) Referring to FIG. **4** and FIG. **5**. FIG. **4** and FIG. **5** are schematic diagrams showing a portion of the optical device **10** with the beam shape modifier **17** according to some embodiments of the present disclosure. In some embodiments, the beam shape modifier **17** is disposed on the top surface t**1** and between two of the first channel filters **122**. The beam shape modifier **17** is configured to receive the light beam from one of the second filters **13** through the vertical descending enabled by the optical signal router **14** and transmission through the second portion **112** of the tray **11**, and reflect the light beam to another one of the second filters **13** along a reverse path and continue the multiplexing or demultiplexing process.
(27) The beam shape modifier **17** is devised to correct beam shape after a predetermined length of propagation. When a light beam transmitted over a predetermined length of propagation in the optical device **10**, the light beam generates unwanted divergence and/or tilting error due to non-ideal factors of the optical components along the path, e.g., filters, optical signal router **14**, and the like. For example, one beam shape modifier **17** is devised after the light beam propagates through **7** collimators on the top surfaces t**1** and t**2**, respectively. The number of collimators that the light beam propagates through can be determined by specific conditions and can be a design factor that optimizes the beam shape correction effect.
(28) In some embodiments, the beam shape modifier **17** includes a microlens having a flat surface **171** on one side and a curved surface **172** on the opposing side. The microlens with a flat surface **171** feature can utilize semiconductor manufacturing facilities and the matured photolithography operations to achieve. The microlens produced by semiconductor photolithography operations can be miniatured in size and well controlled in the curvature of the curved surface **172**.
(29) Referring to FIG. **4**, in some embodiments, the curved surface **172** of the beam shape modifier **17** includes a concave surface, where a distance between the reflection point on the curved surface **172** and the flat surface **171** is smaller than a maximum thickness of the beam shape modifier **17**. The light beam enters the beam shape modifier **17** from the concave surface, reflected by the flat surface **171**, and exits the beam shape modifier **17** from the concave surface. The two passages of the light beam at the concave surface provide an equivalent optical effect as a concave lens, where both sides of the lens are having a concave profile. In some embodiments, the flat surface **171** can be coated with a high-reflective layer to enhance the reflectivity. As shown in FIG. **4**, in some embodiments, the beam shape modifier **17** is disposed only on the top surface t**1** at the first portion **111** of the tray **11**, between adjacent first channel collimators **152** along Y direction, and one beam shape modifier **17** is disposed after the light beam propagates through N collimators **15** and **16**, where N can be any suitable integers. For example, N can be 5, 6, 7, 8, or 9.
(30) Referring to FIG. **5**, in some embodiments, the curved surface **172** of the beam shape modifier **17** includes a convex surface, where a distance between the reflection point on the curved surface **172** and the flat surface **171** is greater than a minimum thickness of the beam shape modifier **17**. The light beam enters the beam shape modifier **17** from the convex surface, reflected by the flat surface **171**, and exits the beam shape modifier **17** from the convex surface. The two passages of the light beam at the convex surface provide an equivalent optical effect as a convex lens, where both sides of the lens are having a convex profile. In some embodiments, the flat surface **171** can be coated with a high-reflective layer to enhance the reflectivity. As shown in FIG. **5**, in some embodiments, the beam shape modifier **17**A is disposed on the top surface t**1** at the first portion **111** of the tray **11**, and the beam shape modifier **17**B is disposed on the top surface t**2** at the second portion **112** of the tray **111**. The beam shape modifier **17**A can be disposed between adjacent first channel collimators **152** along Y direction, and the beam shape modifier **17**B can be disposed between adjacent second channel collimators **16** along Y direction. For the purpose of demonstration, light beam exits the beam shape modifier **17**B, propagating through collimator **16** on the top surface t**2** and the collimator **152** on the top surface t**1**, and enters the beam shape modifier **17**A. However, generally speaking, the light beam should exit one beam shape modifier, propagate through N collimators, where N can be any suitable integers. For example, N can be 5, 6,

7, 8, or 9, and then enter the next beam shape modifier.

(31) Referring back to FIG. **1**, the optical fibers of the first collimators **15** and the second collimators **16** extend along a same direction, i.e., extending away from the optical signal router **14**. Specifically, some of the optical fibers of the second collimators **16** extend directly over the first filters **12** and the beam shape modifiers **17** on the top surface t**1** of the tray **11**.

(32) Referring to FIG. **6**. FIG. **6** is a schematic diagram of an optical collimator structure **70** from a side view perspective according to some embodiments of the present disclosure. The optical collimator structure **60** can represent any one of the first collimators **15** and the second collimators **16** previously discussed.

(33) The optical collimator structure **60** includes a ferrule **61**, an optical fiber **62**, and a lens portion **63**. The ferrule **61** coaxially surrounds and contacts the optical fiber **62**. The lens portion **63** is aligned and optically coupled to the optical fiber **62**. A terminal of the optical fiber **62** and a terminal of the ferrule **61** form a planar surface **611** facing a terminal surface **631** of the lens portion **63**. The ferrule **61** and the lens portion **63** can be fixed or connected through epoxy-containing material or a spacer proximal to the circumferences of the terminal surface **631** of the lens portion **63**.

(34) In some embodiments, the optical collimator structure **60** includes a gap G of free air between the planar surface **611** and the terminal surface **631**. In other embodiments, the optical collimator structure **60** includes a gap G between the planar surface **611** and the terminal surface **631**, which is filled by epoxy containing materials.

(35) In some embodiments, an angle θ**1** between the planar surface **611** and an axial direction of the optical fiber **62** is less than 90 degrees. In some embodiments, the planar surface **611** is substantially parallel to the terminal surface **631**. Therefore, an angle θ**2** between the terminal surface **631** and an axial direction of the optical fiber **62** is substantially equal to the angle θ**1**. In some embodiments, the angle θ**1** about 82 degrees.

(36) As illustrated in FIG. **6**, an entirety of the lens portion **63** is outside of the ferrule **61**. In other words, the ferrule **61** does not surround any portion of the cylindrical surface **632** of the lens portion **63**. Because the lens portion **63** is a free standing component and independent from the ferrule **61**, the lens portion **63** is flexible to be rotated or shifted in respect to the ferrule **61** in order to obtain a Pointing Angle less than 0.05 degree. Aligning and adjusting each of the collimator structures on a DWDM to obtain substantially identical Pointing Angles, i.e., all the Pointing Angles are on a same surface in the free space and tilting toward a same direction, is one of the important aspects in terms of the characters and performance of the DWDM.

(37) Referring to FIG. **7**A and FIG. **7**B. FIG. **7**A and FIG. **7**B are schematic diagrams of the optical collimator structure **60** from a perspective traversing a longitudinal direction of the optical fiber **62** (hereinafter the “cross sectional perspective”) according to some embodiments of the present disclosure. In some embodiments, the optical collimator structure **60** has a square contour from the cross sectional perspective, as illustrated in FIG. **7**A. In other embodiments, the optical collimator structure **60** has a contour including a flat bottom connecting to a curved surface from the cross sectional perspective, as illustrated in FIG. **7**B.

(38) In FIG. **7**A and FIG. **7**B, the ferrule **61** has a flat bottom surface **601** (**601**′) configured to be disposed on or in contact with the tray **11** (not shown). When the optical collimator structure **60** is used as the first collimators **15** described previously, the flat bottom surface **801** (**801**′) is disposed on the top surface t**1**. When the optical collimator structure **60** is used as the second collimators **16**, the flat bottom surface **801** (**801**′) is disposed on the top surface t**2**. The optical collimator structure **70** and the tray **11** have a planar contact in view of the flat bottom surface **601** (**601**′), and the optical collimator structure **60** can be assembled on the tray **11** without extra wedge components of supporting members. Compared to conventional collimator structures with a cylindrical ferrule, the optical collimator structure **60** in the present disclosure provides a steady stand when assembling on the tray **11**. In addition, compared to conventional collimator structures with an extra case

enclosing the ferrule and the lens portion, the optical collimator structure **60** in the present disclosure provides more flexibility in Pointing Angle adjustment during the assembling phase.
(39) Some embodiments of the present disclosure provide an optical device including a tray with a step structure, first filters, second filters, and an optical signal router. The step structure has a first portion and a second portion laterally connected to the first portion. The first portion has a first bottom surface and a first top surface. The second portion has a second bottom surface and a second top surface. The first bottom surface and the second bottom surface are substantially coplanar, and the first portion is thinner than the second portion. The first filters are mounted on the first top surface. The second filters are mounted on the second top surface. The optical signal router optically couples to the first filters and the second filters, and is configured to receive a light beam, transmissible to the tray, from one of the first filters or the second filters.
(40) Some embodiments of the present disclosure provide an optical collimator structure including an optical fiber and a ferrule. The ferrule coaxially surrounds the optical fiber, and is contact with the optical fiber. The ferrule includes a flat bottom surface that enables the optical collimator structure to be disposed on a tray with planar contact.
(41) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other operations and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.
(42) Moreover, the scope of the present application is not intended to be limited to the particular embodiments of the process, machine, manufacture, composition of matter, means, methods and steps described in the specification. As one of ordinary skill in the art will readily appreciate from the disclosure of the present invention, processes, machines, manufacture, compositions of matter, means, methods, or steps, presently existing or later to be developed, that perform substantially the same function or achieve substantially the same result as the corresponding embodiments described herein may be utilized according to the present invention. Accordingly, the appended claims are intended to include within their scope such processes, machines, manufacture, compositions of matter, means, methods, or steps.

Claims

1. An optical device, comprising: a tray with a step structure, the step structure having a first portion and a second portion laterally connected to the first portion, the first portion having a first bottom surface and a first top surface, the second portion having a second bottom surface and a second top surface, the first bottom surface and the second bottom surface being substantially coplanar, and a first thickness of the first portion is less than a second thickness of the second portion; a plurality of first filters mounted on the first top surface; a plurality of second filters mounted on the second top surface; and an optical signal router optically coupling to the first filters and the second filters, the optical signal router being configured to receive a light beam, through the second portion of the tray, from one of the first filters and guide the light beam to one of the second filters, and the optical signal router being further configured to receive the light beam from the one of the second filters and guide the light beam, through the second portion of the tray, to another one of the first filters.
2. The optical device of claim 1, wherein the optical signal router is a right angle prism vertically offsetting the light beam from the first filters or the second filters.
3. The optical device of claim 1, further comprising: a plurality of first collimators mounted on the

first top surface, each optically coupling to the respective first filters; and a plurality of second collimators mounted on the second top surface, each optically coupling to the respective second filters.

4. The optical device of claim 3, wherein one of the first collimators comprises: a first optical fiber extending over the first portion of the step structure; and a first ferrule coaxially surrounding and in contact with the first optical fiber; wherein the first ferrule having a flat bottom surface disposed on the first top surface.

5. The optical device of claim 4, wherein the first collimator further comprises: a lens portion aligned and optically coupled to the first optical fiber, wherein a terminal of the first optical fiber and a terminal of the first ferrule form a planar surface facing a terminal surface of the lens portion.

6. The optical device of claim 5, wherein an entirety of the lens portion is outside of the first ferrule.

7. The optical device of claim 4, wherein the first ferrule comprises a square contour from a perspective traversing the first optical fiber.

8. The optical device of claim 3, wherein one of the second collimators comprises: a second optical fiber extending over the first portion and the second portion of the step structure; and a second ferrule coaxially surrounding and in contact with the second optical fiber, wherein the second ferrule having a flat bottom surface disposed on the second top surface.

9. The optical device of claim 8, further comprising: a first beam shape modifier disposed on the first top surface and between two of first filters, the first beam shape modifier being configured to receive the light beam from the one of the second filters and reflect the light beam to another one of the second filters on the second top surface.

10. The optical device of claim 9, wherein the first beam shape modifier comprises a microlens having a flat surface with reflection coating.

11. The optical device of claim 10, wherein the first beam shape modifier further comprises a concave or convex surface opposite to the flat surface.

12. The optical device of claim 9, wherein the second optical fiber extends directly over the first beam shape modifier on the first top surface.

13. The optical device of claim 9, further comprising: a second beam shape modifier disposed on the second top surface and between two of second filters, the second beam shape modifier being configured to receive the light beam from the one of the first filters and reflect the light beam to the another one of the first filters on the first top surface.

14. The optical device of claim 4, wherein the flat bottom surface enables the one of the first collimators to be disposed on a tray with planar contact.

15. The optical device of claim 5, wherein an angle between the planar surface and an axial direction of the optical fiber is less than 90 degrees.

16. The optical device of claim 5, wherein the terminal surface of the lens portion is parallel to the planar surface of the optical fiber and the ferrule.

17. The optical device of claim 5, wherein the one of the first collimator further comprises a gap between the planar surface and the terminal surface.

18. The optical device of claim 17, wherein the gap is a gap of free air.

19. The optical device of claim 17, wherein the gap is filled with epoxy-containing materials.

20. The optical device of claim 4, wherein the first ferrule comprises a contour including a flat bottom connecting to a curved surface from a perspective traversing the first optical fiber.